

CEO & Leadership Review

Can this product safely become the company's operating system?

EXECUTIVE VERDICT

CONTROLLED INTERNAL BETA ONLY. Do not approve a full FieldRoutes replacement or public commerce launch until the P0 controls are closed and proven. The prototype represents most of the lifecycle, but leadership cannot yet trust lead capture, activation, cash reporting, cancellation, capacity, or employee permissions.

Prepared for	Review date	Scope
CEO, executive team, functional leaders, and board-level reviewers	July 15, 2026	Business logic, operating controls, customer trust, and low-skill usability

This review evaluates whether each feature helps the business complete a reliable customer transaction. It is intentionally not a code-quality review. The standard is operational truth: the system should guide the employee, enforce prerequisites, complete all related business effects, and visibly route exceptions.

Review method and scope

Three independent reviews examined the product from the perspectives below, then reconciled their findings into launch gates and operating requirements. The local product source, data model, workflows, business copy, and integrations were inspected. The available hosted deployment could not be opened in the review environment, so this report does not claim a visual usability test of the running production interface.

- CEO: growth, cash, margin, risk, scalability, and reliability of management reporting.
- Leadership team: policy ownership, role separation, KPI definitions, approval thresholds, and rollout risk.
- Cross-functional lifecycle: acquisition through retention, including every handoff that creates or loses business value.

The strongest finding is the consistency of the conclusions: leadership, customer, and field perspectives all identified the same failure class - a user can complete an apparent action while billing, scheduling, communication, or downstream records remain unfinished or contradictory.

What is strategically strong

- The core entities - customer, plan, job, report, invoice, agreement, quote, and portal - form the right foundation for a service-business operating system.
- Deterministic pricing code is preferable to allowing AI to invent prices; AI can extract facts while an approved rule set computes the offer.
- Plan templates, e-signature evidence, signed PDFs, service reports, GPS, product logs, and customer document access can standardize execution and reduce disputes.
- Stripe payment methods, subscriptions, webhooks, and several idempotency protections are useful transaction foundations.
- Customer groups and multi-property access support the valuable HOA/property-management growth motion.
- Reminders, report delivery, review requests, and recurring-job creation can remove manual office work once failures are controlled.

What leadership cannot currently trust

Leadership promise	Current business exposure
Lead capture	Core public forms still create records in FieldRoutes while the CRM tells staff that website inquiries appear in CRM.
Price integrity	Legacy residential pricing and the new CRM rate card disagree; several pricing paths can quote different amounts.
Activation	Signed or manually converted customers can become ACTIVE without a payment method, first job, subscription, or next action.
Cash reporting	Prepaid web bookings can move money without an Invoice; dashboard paid logic uses issue date, and refunds are incomplete.
Service-to-cash	A one-time job can be completed without an invoice or charge; recurring billing can depend on an employee remembering to start it.
Cancellation	Customer, plan, agreement, Stripe subscription, future jobs, reminders, and portal access can contradict one another.
Capacity	Public availability is not reserved or revalidated at payment, so multiple customers can buy the same capacity.
Governance	The broad OFFICE role can change pricing, agreements, charges, statuses, and privileged access without approval or durable audit.
Exceptions	Several downstream failures are logged or swallowed, while the visible upstream action still looks successful.

P0 launch blockers

Priority	Business requirement	Acceptance condition
P0	One source of truth and one price book	Every public and internal intake creates or reconciles to one customer ID, one lead queue, and one versioned approved price. No duplicate public/legacy price engine remains.
P0	Guarded customer activation	A customer cannot become ready for service until agreement/override, scope, approved price, payment terms, first job, billing start rule, owner, and next action are complete.
P0	Durable exception control tower	Every partial failure creates a visible, owned, due-dated task. Signed-not-converted, active-not-billing, active-no-visit, completed-unbilled, and communication failures are impossible to miss.
P0	Authoritative financial ledger	Every Stripe and offline movement produces a ledger/invoice event. Quoted, booked, invoiced, paid, refunded, fees, write-offs, and net collected reconcile daily.
P0	Coordinated cancellation	One command determines effective date, Stripe action, refund/credit, future jobs, plan/customer state, access, reason, approvals, and customer confirmation.
P0	Atomic capacity claim	Checkout holds and then atomically claims a live skill/window/shift capacity slot before payment is confirmed.
P0	Roles, approvals, and audit	Owner/admin, manager, sales, dispatch, finance, service, tech, and read-only roles replace broad OFFICE authority. Material overrides require reason and approval.
P0	Margin floors	Every quote tests current labor, drive, materials, fees, lead cost, service frequency, and expected callbacks. Below-floor prices require manager approval.

Canonical lifecycle leadership should approve

State	Entry proof	Owner	Required next result
1. Inquiry	Durably captured, deduplicated, attributed	Sales	New lead SLA starts
2. Qualified	Service area, offer type, scope, authority, and risk known	Sales	Price path is selected
3. Quoted	Approved scope, terms, price version, margin, and expiration	Sales/manager	Follow-up is scheduled
4. Signed pending setup	Customer accepted; operational prerequisites still explicit	Customer service	Payment and first service setup
5. Ready to schedule	Agreement, payment terms, scope, price, and site profile complete	Operations	Capacity-constrained appointment
6. Active servicing	Service and billing clocks are independently healthy	Operations/finance	Deliver, bill, report, recur
7. Exception	A promise failed or prerequisite is missing	Named owner	Resolve by due time or escalate
8. Paused/canceled	All downstream consequences reconciled	Authorized role	Final billing, service, access, and retention outcome

Feature-by-feature leadership disposition

Feature	Verdict	Leadership business requirement
Authentication / magic links	Keep with controls	Low friction is valuable; split privileges, control invitations, and add revocation/audit.

Feature	Verdict	Leadership business requirement
Dashboard	Rebuild before CEO reliance	Useful office snapshot, but not a trustworthy revenue, cash, margin, capacity, or exception view.
Leads	Redesign	Website demand must land here; add owner, stage, SLA, next action, attempts, consent, value, and loss reason.
AI pricing	Keep with guardrails	Use AI for fact extraction; approved deterministic rules must set price and validate margin.
Market rates	Research only	Competitor-derived prices require human approval, effective dates, bounds, and margin checks before sale.
Pricing outcome log	Automate	Won/lost results should derive from quotes and agreements; require loss reasons and economic context.
Plan templates	Keep and govern	Version, effective date, owner, legal approval, scope, exclusions, and retirement rules are required.
Quotes / e-sign	Keep; harden	Add expiry/version/scope/authority; voiding must cascade; signing must launch guarded activation.
Customers	Keep; expand lifecycle	Add identity matching, owner, lifecycle reasons, merge, and explicit service/billing readiness.
Customer groups	Strategic	Add account economics, master terms, renewal, owner, billing hierarchy, and scoped access.
Service plans	Core; redesign state	Separate entitlement, service cadence, billing cadence, delivery health, pause, and cancellation.
Payments / Stripe	Keep; complete ledger	Add dunning, refunds, reconciliation, fees, evidence, approval, and customer receipts.
Manual billing	Finance escape hatch	Limit by role and threshold; require reason, supporting reference, approval, and audit.
Jobs	Core; add prerequisites	Every work order needs scope, price/billing disposition, duration, site readiness, and due/promise dates.
Schedule / routes	Keep foundation	Add skill, capacity, shift, PTO, territory, drive, time-window, and over-capacity controls.
Technician workflow	Keep narrow; enforce	Row-scope to assigned work and require service-specific evidence and valid transitions.
Reports / GPS / photos	High trust value	Add QA, immutability, delivery state, location exceptions, incident and callback linkage.
Product catalog	Keep; make compliant	Approval, cost, inventory, label/SDS, certification, and usage constraints are necessary.
Portal	Keep; increase deflection	Add pay, retry, reschedule, cancel, request service, update access/safety details, and support.
Communications	Keep; make truthful	Consent, delivery status, retry, acknowledgement, frequency caps, and exception escalation are required.
Legacy public forms	Migrate or formally sync	They cannot remain a shadow acquisition and pricing system.
Public booking	Delay launch	Expose only after live reservation, ledger, cancellation, offer eligibility, and customer identity are safe.
PWA / install	Useful, not sufficient	A cached shell is not field resilience; durable offline data and queued writes are required.

Minimum leadership scorecard

Domain	Required measures
Revenue	Booked, invoiced, collected, net collected, recognized, MRR/ARR, new/lost MRR, average ticket
Profit	Contribution margin by plan, service, customer, source, geography, and technician
Growth	Leads, qualified rate, quote rate, close rate, sales-cycle time, CAC and payback by source
Retention	Logo/revenue churn, pause, cancellation reasons, tenure, save rate, reactivation
Cash	A/R aging, DSO, failed-payment recovery, refunds, write-offs, processor fees, reconciliation exceptions
Operations	Due/unscheduled/overdue visits, capacity, route utilization, drive ratio, on-time rate, report compliance
Quality	Callbacks, free re-treatments, complaints, first-time resolution, CSAT/NPS, review conversion
Risk	Unsigned/expired agreements, licenses/certifications, product exceptions, privileged changes, unresolved incidents

Every metric needs a definition, target, prior-period comparison, trend, owner, and drill-down to the underlying records.

Leadership decisions engineering must not invent

1. What ACTIVE means separately for customer relationship, legal agreement, service entitlement, delivery, and billing.
2. Which system is authoritative during FieldRoutes migration, and the deadline for eliminating dual entry and dual pricing.
3. When recurring billing starts for each offer and what happens when the trigger fails.
4. The approved cancellation, rescheduling, no-access, refund, credit, weather, and satisfaction matrix by product/channel.
5. Minimum contribution margin and who can approve price, discount, fee, or term exceptions.
6. Which jobs may be performed without signed terms or payment readiness, and the evidence/approval required.
7. Role definitions and approval thresholds for price, refunds, write-offs, offline payments, plan changes, agreement voids, and privileged invitations.
8. Contract cadence versus service completion cadence, allowed delivery tolerance, and customer-notification rules.
9. Data retention, audit, incident, compliance, and report-amendment policy.
10. The phased cutover and rollback criteria for replacing FieldRoutes.

Recommended build and rollout order

Phase	Exit requirement
0. Policy sign-off	Name the systems of record, state definitions, price book, margin floor, billing triggers, cancellation matrix, roles, and SLAs.
1. Control foundation	Implement roles, approvals, audit events, exception objects, ownership, and authoritative money events.
2. Lifecycle transactions	Build atomic activation, service completion/billing disposition, cancellation, and customer/tech deactivation workflows.
3. Funnel migration	Move every public form into the canonical CRM path; dedupe and reconcile existing FieldRoutes records.
4. Internal beta	Use experienced staff, no public self-booking, daily reconciliation, and zero unresolved P0 exception classes.
5. Controlled field pilot	Small technician cohort with offline, assignment, safety, product, report, billing, and dispatch controls proven.
6. Customer commerce	Enable booking only after live capacity reservation, offer eligibility, receipts, cancellation, and support recovery are proven.
7. KPI scale-up	Build executive analytics after state and ledger definitions are stable and reconciled.

The low-skill-safe operating doctrine

The system decides; the employee confirms.

A frontline employee should see one safe next action, why it is required, and what happens next. The employee should not have to remember a hidden prerequisite, a second screen, or a cleanup step in another record.

Principle	Business rule
One state	Each lead, customer, plan, job, invoice, and exception has one unambiguous business state.
One owner	Every open promise and exception has a named owner, due time, and escalation path.
One primary action	The screen emphasizes the next valid action and suppresses invalid or premature actions.
Complete transactions	Activation, completion, cancellation, and payment update every affected record together.
Guardrails	Price, safety, compliance, scheduling, and financial rules are enforced by the system, not training memory.
Truthful outcomes	The app never says sent, paid, booked, active, or complete unless that exact result is durable.
Manager exceptions	Overrides require a role, reason, evidence where appropriate, and an immutable audit event.
Recoverable failure	A partial failure creates a visible work item; it never disappears into a log or a green success screen.

Selected evidence

Finding	Representative source evidence
Dual systems / pricing	README.md:7-11; apps/web/amplify/functions/lead-intake/handler.ts:532-619; apps/crm/src/office/Leads.tsx:65-68; crm-pricing/rateCards.ts:50-54
Activation and recurring billing	agreement-public/handler.ts:91-124; CustomerDetail.tsx:473-507, 923-947; shared/bookingFinalize.ts:172-189, 291-294
Exceptions	agreement-public/handler.ts:277-288; shared/recurring.ts:39-47, 109-111; Dashboard.tsx:101-119
Financial ledger	Dashboard.tsx:88-95; shared/bookingFinalize.ts:136-266; data/resource.ts:474-500; CustomerDetail.tsx:1131-1149
Cancellation	crm-billing/handler.ts:214-234; CustomerDetail.tsx:425-437, 856-873; booking-public/handler.ts:687-708
Capacity	booking-public/availability.ts:74-87, 138-191; booking-public/handler.ts:475-556
Permissions	data/resource.ts:15-18; More.tsx:101-188; crm-billing/handler.ts:343-383

Evidence paths are relative to the BuzzKill repository. They illustrate the business rule observed; this report is not a code defect inventory.